

Project Statement: CS361L – Computer Architecture Lab

Department of Computer Science & Engineering

CS 5th semester - Fall 2025

1. Introduction

This document defines the project component for the CS361L Computer Architecture & Parallel Programming Laboratory. The purpose of this project is to provide applied exposure to instruction execution models, pipelining behavior, and shared-memory parallelism using OpenMP. One project option must be selected and completed with implementation, analysis, and professionally structured reporting.

2. Project Objectives

The project aims to:

- Analyze processor execution models and their performance implications.
- Compare serial and parallel execution using OpenMP.
- Interpret instruction flow, hazards, and throughput behavior.
- Relate hardware execution to software performance outcomes.
- Produce technically organized documentation and defend implementation decisions.
- Collaborate effectively within a structured engineering workflow.

3. Project Options

One of the following tracks must be selected:

Option 1: OpenMP-Driven Parallel Performance Engineering

Focus: Advanced shared-memory multi-threading and scalability.

Condensed Description:

Select a computational workload and develop serial and OpenMP-parallelized versions. Analyze thread scheduling, synchronization overhead, memory access patterns, and core utilization.

Provide execution timing, speedup graphs, efficiency evaluation, and bottleneck interpretation.

Option 2: Instruction-Level Execution and Microarchitectural Analysis

Focus: Pipeline execution behavior and performance interpretation.

Condensed Description:

Examine execution using Single-Cycle, Multi-Cycle, or 5-Stage Pipelined models. Analyze hazards,

stalling, forwarding, CPI estimation, and dependency effects. Include pipeline diagrams, cycle-accurate traces, and theoretical versus interpreted performance relationships.

Option 3: Hybrid Parallel and Pipeline Interaction Study

Focus: Combined thread-level and instruction-level performance evaluation.

Condensed Description:

Implement a workload using OpenMP while analyzing pipeline effects such as cache behavior, stalls, and dependency interactions. Provide comparative performance models, profiling data, and a synthesized justification of how microarchitectural factors influence parallel throughput.

4. Team Formation Requirements

- A maximum of **three (3)** members per team.
- Contribution must be meaningful and distributed.
- Required member details: Full Name. Registration Number, Institutional Email

5. Proposal Submission Guidelines

Proposal Deadline: This Thursday

- Hard copy and Soft on the teams.
- Late submission results in a deduction of marks.
- Maximum length: **2 pages**
- Required sections:
 1. **Cover Page** – Title, option, member details
 2. **Project Overview** – relevance, problem, outcomes
 3. **Methodology** – approach, tools

Final Project Deadline: 10 December 2025

Submission must be a **proper ZIP file** containing:

- a. **Source code**
- b. Report (PDF)
- c. Supporting material

An incomplete submission results in a deduction of marks.

6. Additional Requirements

- Professional formatting required.
- Low-effort or unclear work will not be accepted.
- Reused, copied, or AI-generated material will lose marks.
- Identical work across groups results in **ZERO**.